PROBLEM STATEMENT #5

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Section: C

Lab 2: Agile Backlog Creation & Sprint Simulation

Scenario: Academic Elective Bidding System
Project Name: Academic Elective Bidding System
Project Key: AEBS

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1. Backlog & Story Point Summary Table

Epic	Story ID	User Story	Points	Priority
Student Elective Bidding	AEBS-6	Submit Elective Preferences	5	High
Student Elective Bidding	AEBS-7	Validate Bidding Credits	3	High
Student Elective Bidding	AEBS-8	Validate Prerequisites	3	High
Bidding Configuration	AEBS-9	Configure Bidding Window	3	High
Bidding Configuration	AEBS-10	Configure Elective Seat Capacity	3	High
Bidding Configuration	AEBS-11	Configure Bidding Rules	5	High
Elective Allocation	AEBS-12	Rank Students by Bid Credits	5	High
Elective Allocation	AEBS-13	Allocate Students to Electives	5	High
Elective Allocation	AEBS-14	Apply CGPA Tie-Breaker	3	High
Allocation Results & Notifications	AEBS-15	View Allocation Result	3	High
Allocation Results & Notifications	AEBS-16	Send Allocation Notifications	3	Medium
Allocation Results & Notifications	AEBS-17	Display Allocation Details	3	Medium
Allocation Override & Audit	AEBS-18	Manual Allocation Override	5	High
Allocation Override & Audit	AEBS-19	Record Allocation Changes	3	Medium
Allocation Override & Audit	AEBS-20	View Allocation Audit Trail	3	Medium

Total Backlog Estimate: **50 Story Points** across 15 User Stories and 5 Epics.

2. Deliverables Breakdown

Deliverable 1 & 2: Backlog, Epics, & Story Point Assignments

Epics Created:

• Epic 1: Student Elective Bidding (AEBS-1) – Enables students to submit ranked elective preferences and validates their eligibility to bid.

• Epic 2: Bidding Configuration (AEBS-2) – Allows administrators to configure the bidding window, seat capacity, and bidding rules.

• Epic 3: Elective Allocation (AEBS-3) – Handles ranking and allocation of students to electives based on bid credits and CGPA tie-breakers.

• Epic 4: Allocation Results & Notifications (AEBS-4) – Displays allocation results to students and sends completion notifications.

• Epic 5: Allocation Override & Audit (AEBS-5) – Supports manual allocation overrides and maintains a full audit trail of changes.

User Stories & Fibonacci Estimates:

Story 1.1: Submit Elective Preferences (AEBS-6 – 5 points, High Priority)

As a student, I want to submit my ranked elective preferences, so that I can be considered for allocation based on my choices.

Story 1.2: Validate Bidding Credits (AEBS-7 – 3 points, High Priority)

As a student, I want the system to validate my available bidding credits before submission, so that I cannot bid more than what I have.

Story 1.3: Validate Prerequisites (AEBS-8 – 3 points, High Priority)

As a student, I want the system to check that I meet the prerequisites for an elective, so that I only bid for electives I am eligible to take.

Story 2.1: Configure Bidding Window (AEBS-9 – 3 points, High Priority)

As an admin, I want to set the start and end dates of the bidding window, so that students can only submit bids within the defined period.

Story 2.2: Configure Elective Seat Capacity (AEBS-10 – 3 points, High Priority)

As an admin, I want to define the seat capacity for each elective, so that allocation respects the available slots.

Story 2.3: Configure Bidding Rules (AEBS-11 – 5 points, High Priority)

As an admin, I want to configure the rules governing the bidding process (maximum preferences, credit limits), so that bidding runs consistently and fairly.

Story 3.1: Rank Students by Bid Credits (AEBS-12 – 5 points, High Priority)

As the system, I want to rank students by the bid credits allocated to each elective, so that higher-priority bids are considered first during allocation.

Story 3.2: Allocate Students to Electives (AEBS-13 – 5 points, High Priority)

As the system, I want to allocate students to electives based on their ranked bids and available seats, so that seats are filled fairly and efficiently.

Story 3.3: Apply CGPA Tie-Breaker (AEBS-14 – 3 points, High Priority)

As the system, I want to use CGPA as a tie-breaker when bid credits are equal, so that allocation remains fair when demand exceeds supply.

Story 4.1: View Allocation Result (AEBS-15 – 3 points, High Priority)

As a student, I want to view my elective allocation result, so that I know which elective I have been assigned to.

Story 4.2: Send Allocation Notifications (AEBS-16 – 3 points, Medium Priority)

As a student, I want to receive a notification once allocation is complete, so that I am promptly informed of my result.

Story 4.3: Display Allocation Details (AEBS-17 – 3 points, Medium Priority)

As a student, I want to see detailed information about my allocated elective (timing, faculty, credits), so that I can plan my schedule accordingly.

Story 5.1: Manual Allocation Override (AEBS-18 – 5 points, High Priority)

As an admin, I want to manually override a student's allocation when needed, so that exceptional cases can be handled fairly.

Story 5.2: Record Allocation Changes (AEBS-19 – 3 points, Medium Priority)

As an admin, I want every manual change to be recorded, so that there is a traceable history of allocation decisions.

Story 5.3: View Allocation Audit Trail (AEBS-20 – 3 points, Medium Priority)

As an admin, I want to view the audit trail of all allocation overrides and changes, so that I can ensure accountability and transparency.

Backlog Screenshots (Jira):

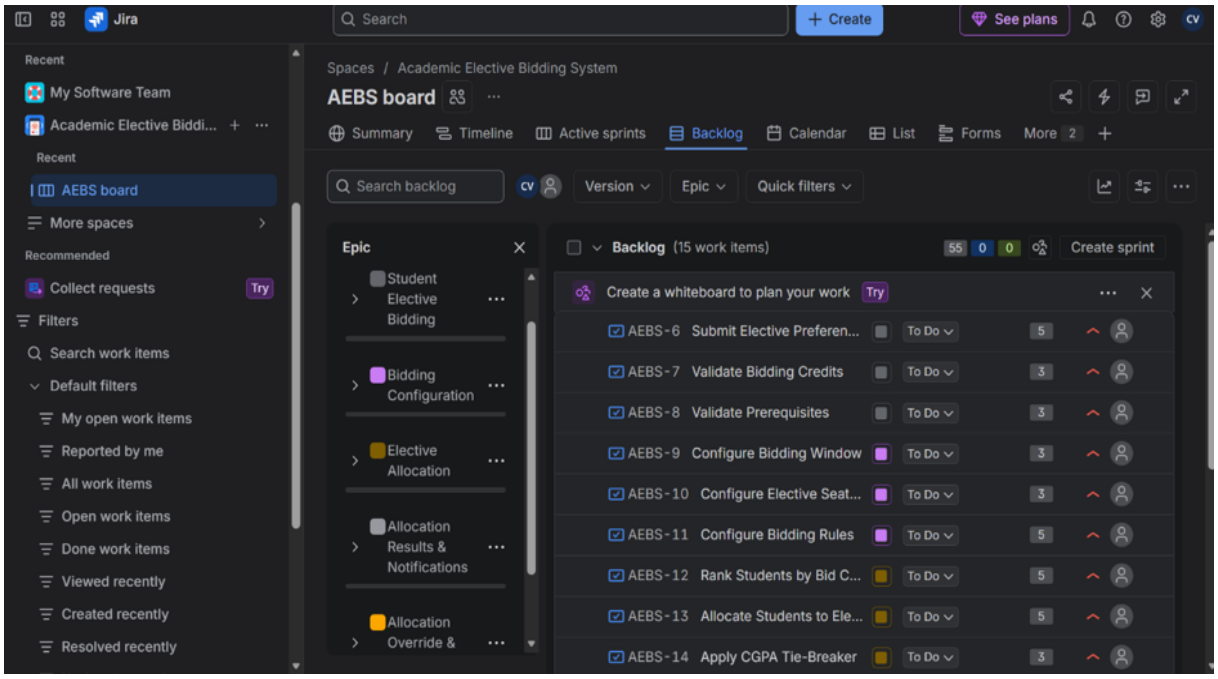


Fig 1: AEBS Backlog showing Epics and the first set of User Stories.

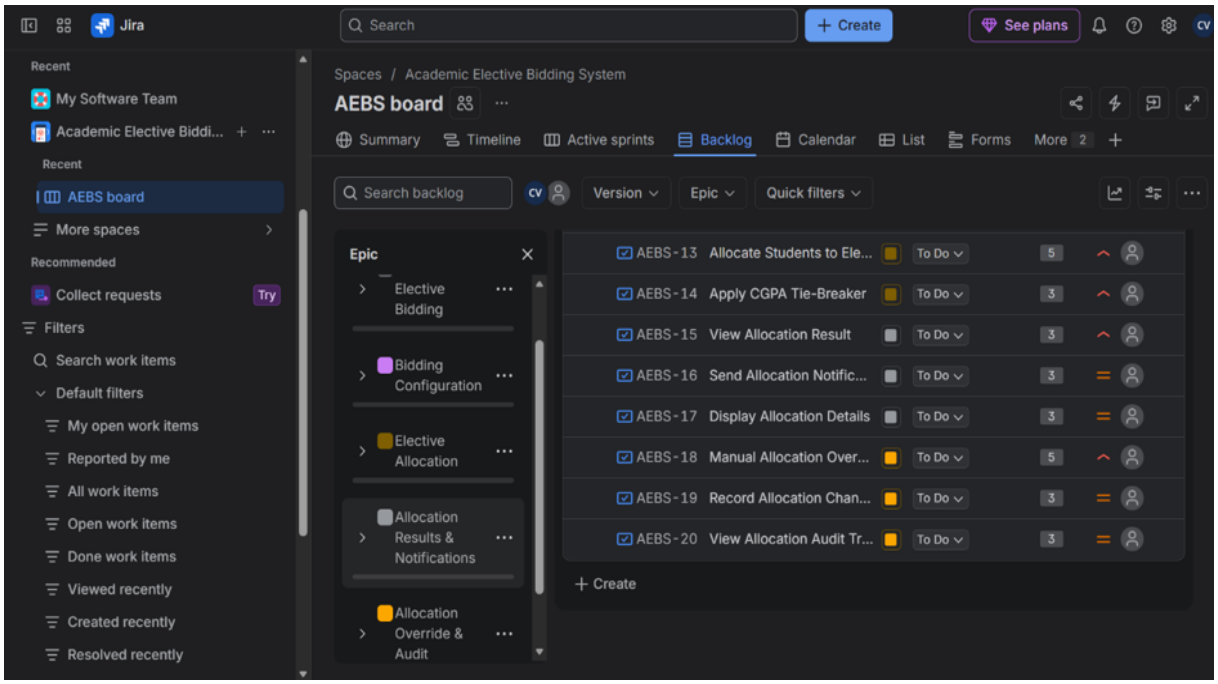


Fig 2: AEBS Backlog continued, showing remaining User Stories with story point estimates.

Deliverable 3: Sprint Board (Active Sprint View)

Three sprints were planned and executed for the AEBS project:

Sprint	Work Items	Committed Capacity
AEBS Sprint 1	AEBS-6, AEBS-7, AEBS-8, AEBS-9, AEBS-10, AEBS-12	22 Story Points
AEBS Sprint 2	AEBS-13, AEBS-14, AEBS-15, AEBS-16, AEBS-17	17 Story Points
AEBS Sprint 3	AEBS-18, AEBS-19, AEBS-20	11 Story Points

Sprint Name: AEBS Sprint 3
Committed Capacity: 11 Story Points (AEBS-18, AEBS-19, AEBS-20)
Execution: All three work items in Allocation Override & Audit were moved through the workflow columns (To Do → In Progress → Done) during the simulated sprint timebox, closing out the AEBS backlog.

Sprint 1

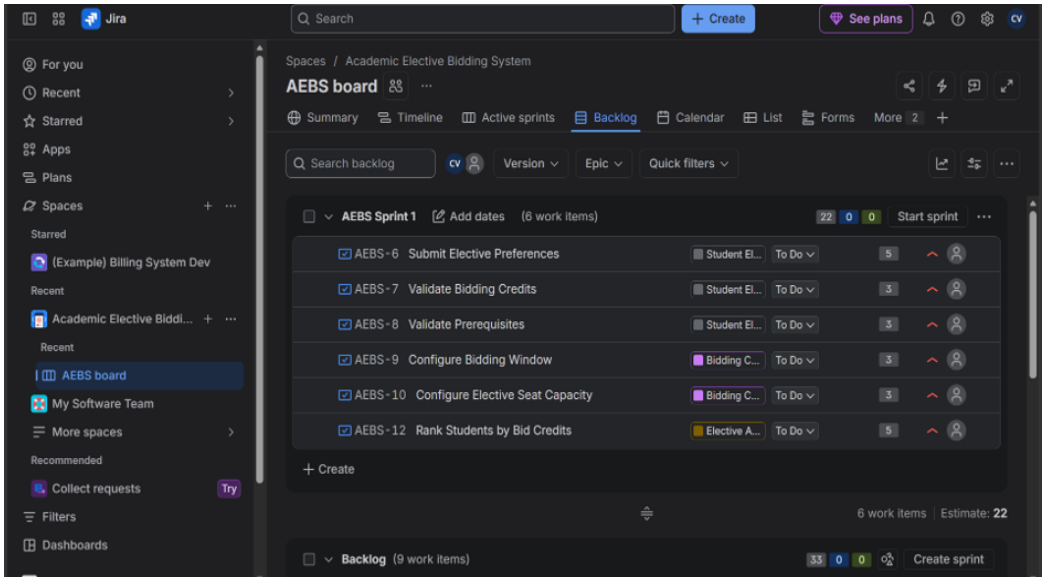


Fig 3: AEBS Sprint 1 started with 6 committed work items (22 points), all in To Do.

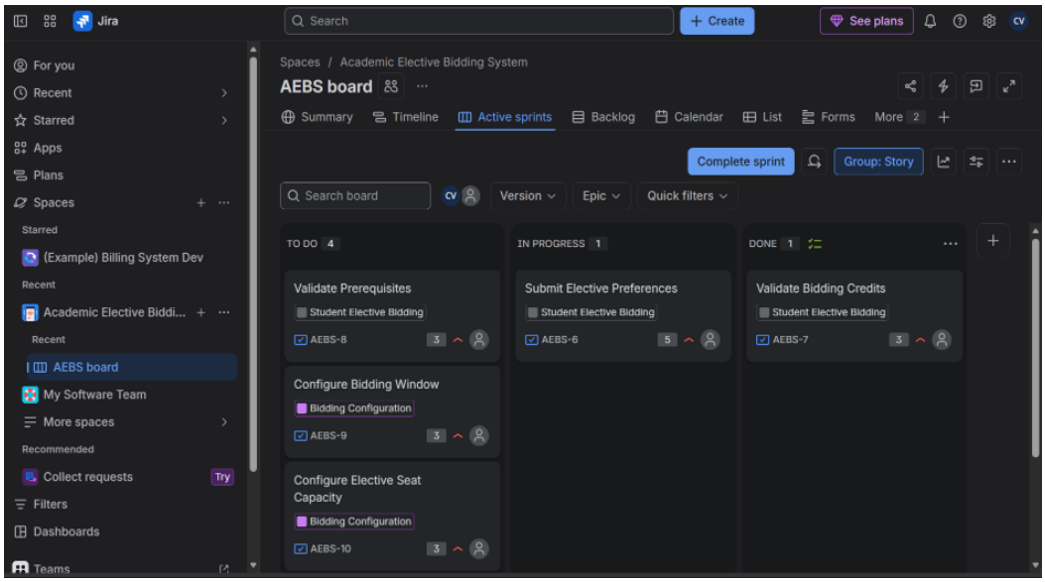


Fig 4: AEBS Sprint 1 in progress – items moving across To Do, In Progress, and Done.

Sprint 2

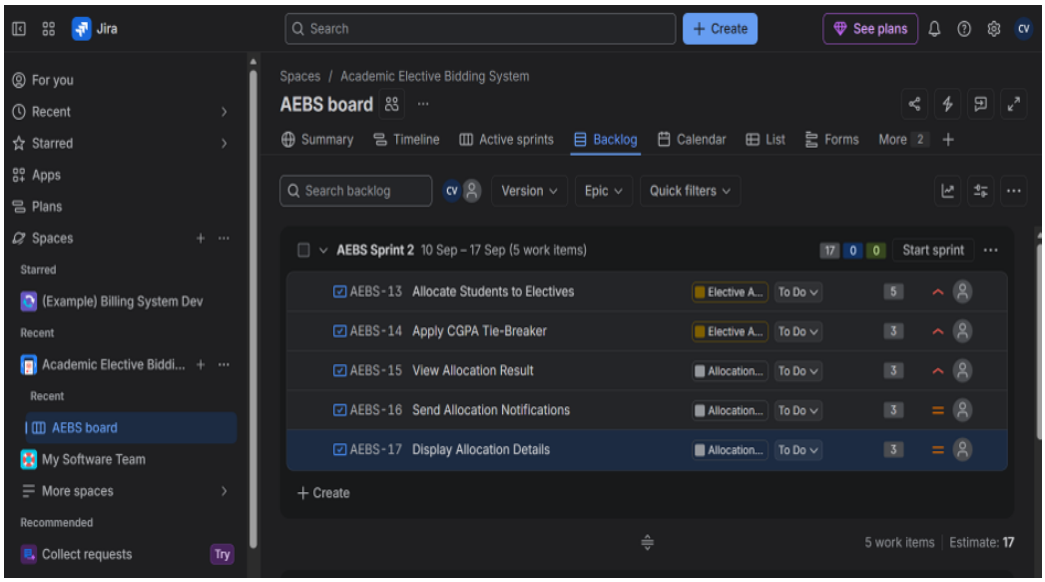


Fig 5: AEBS Sprint 2 started with 5 committed work items (17 points), all in To Do.

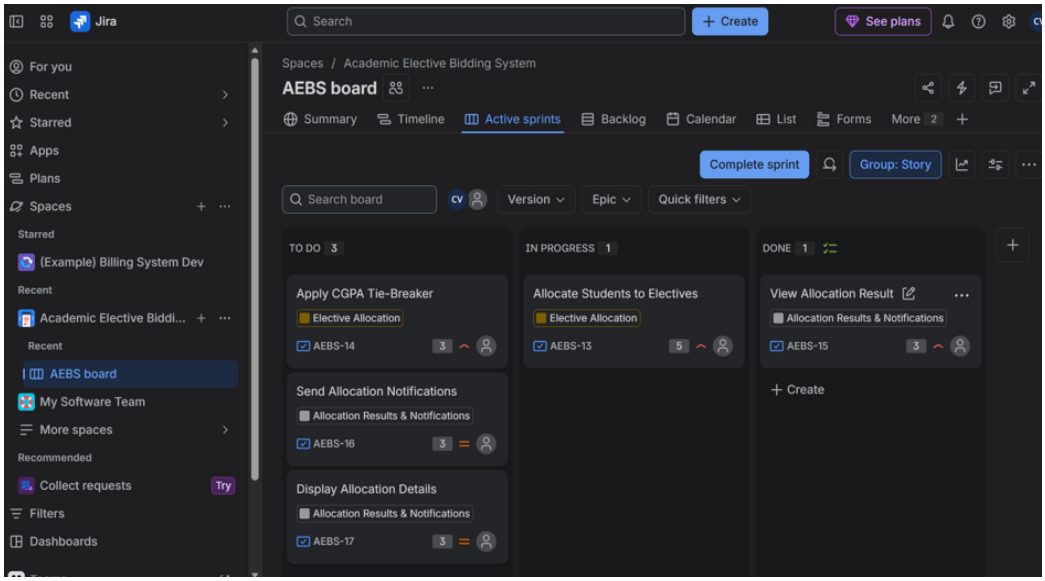


Fig 6: AEBS Sprint 2 in progress – items distributed across To Do, In Progress, and Done.

Sprint 3

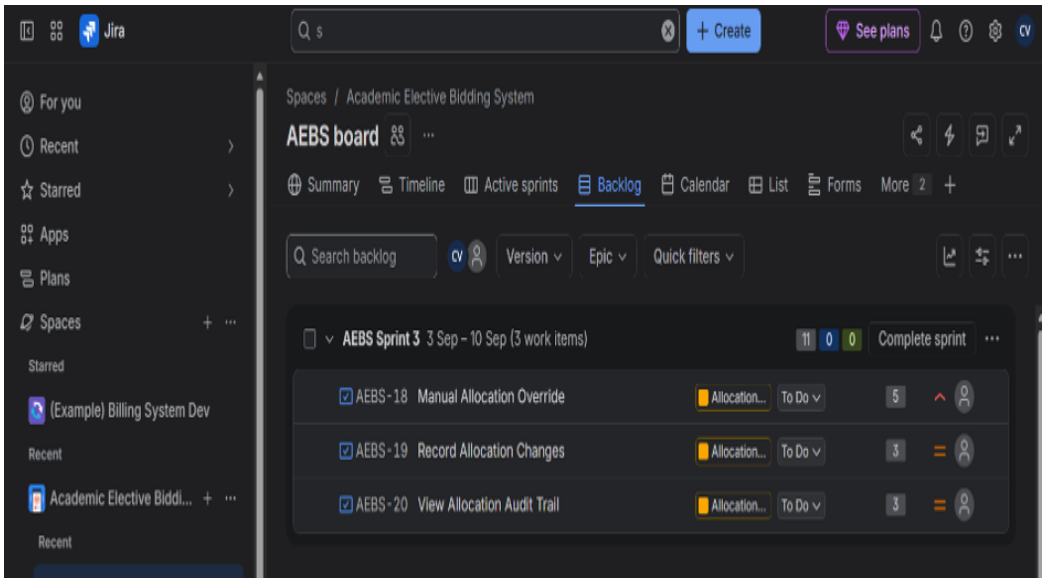


Fig 7: AEBS Sprint 3 started with 3 committed work items (11 points), all in To Do.

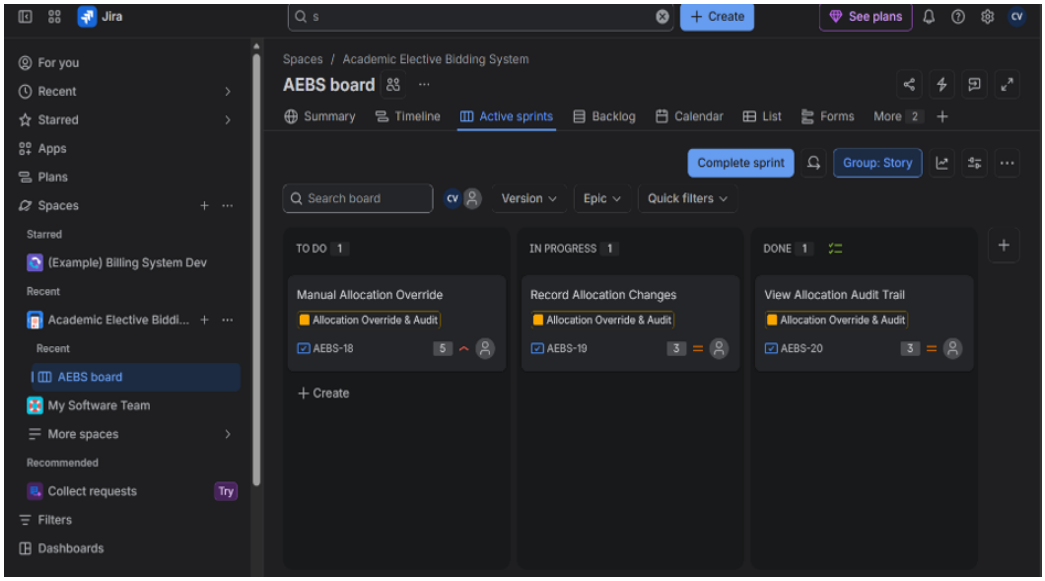


Fig 8: AEBS Sprint 3 in progress – one item In Progress, one item Done, closing out the AEBS backlog.

Deliverable 4: Burndown Chart & Event Log

Sprint Status: Completed
Start Value: 10 Story Points remaining at sprint start
Remaining Value at Sprint Close: 0 Story Points
Observation: The burndown line for AEBS Sprint 3 drops steadily from 10 remaining story points on Sep 4 to 0 by Sep 10, tracking close to the ideal guideline line as AEBS-18 (Manual Allocation Override, 5 pts), AEBS-19 (Record Allocation Changes, 3 pts), and AEBS-20 (View Allocation Audit Trail, 3 pts) were resolved across the sprint, indicating steady and predictable progress with no scope creep.

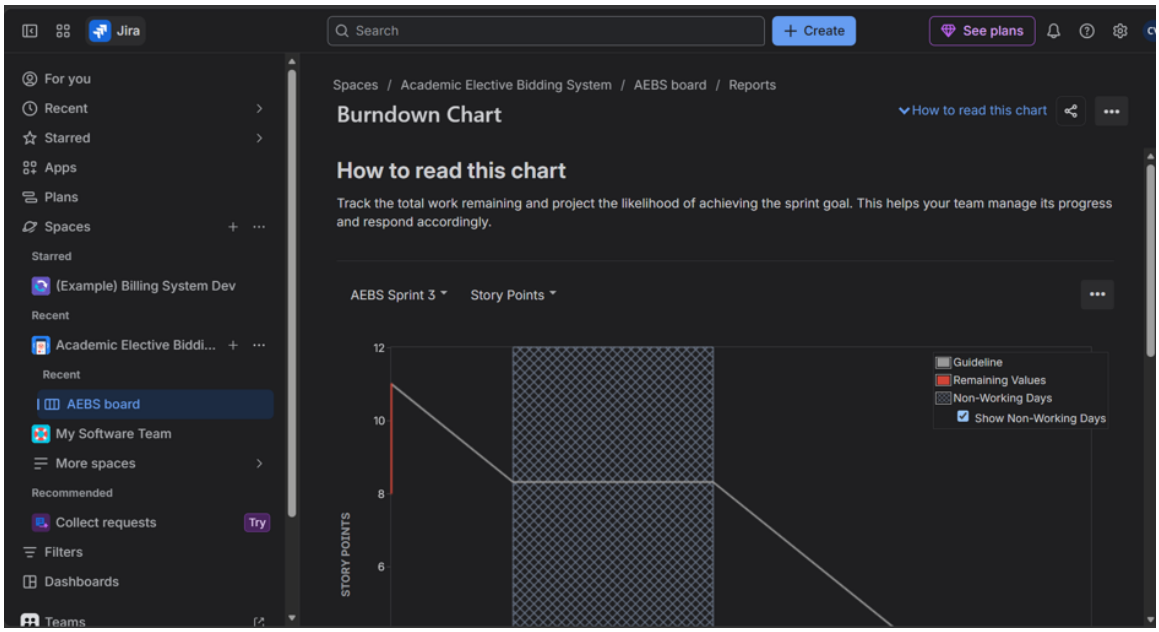


Fig 9: Burndown Chart for AEBS Sprint 3 shortly after sprint start.

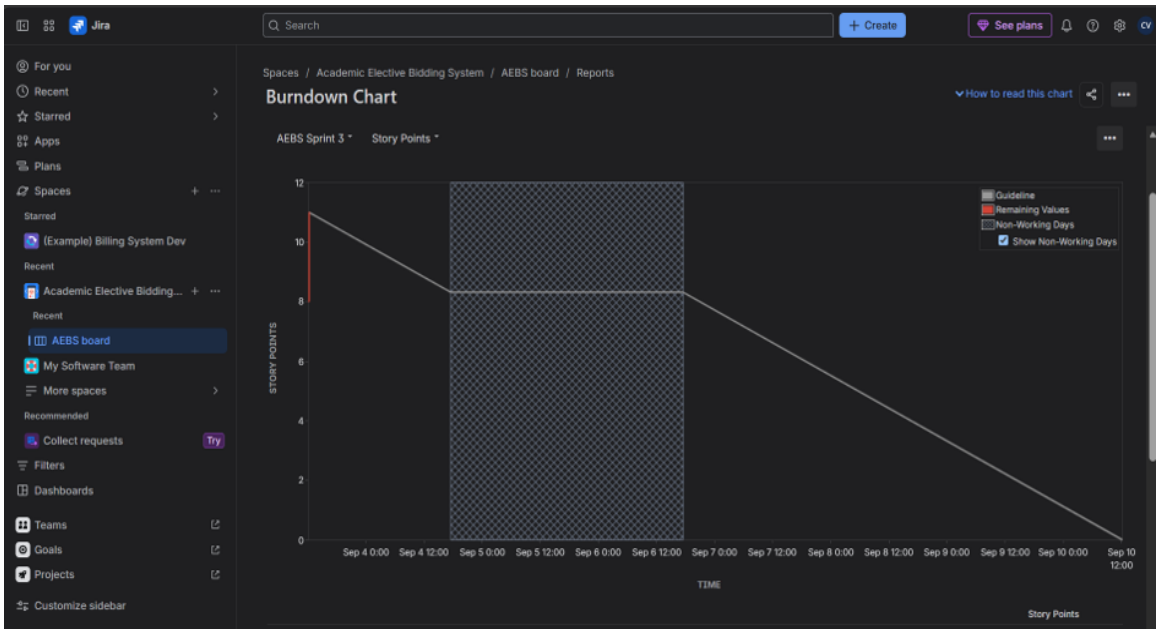


Fig 10: Completed Burndown Chart for AEBS Sprint 3, showing story points reaching zero by Sep 10.

3. Reflection Questions & Analysis

1. Did your estimations reflect the actual effort?

Ans. Yes. Simple configuration tasks like Configure Bidding Window (AEBS-9) and Configure Elective Seat Capacity (AEBS-10) were estimated at 3 points each, since they involve straightforward form-based settings. In contrast, Allocate Students to Electives (AEBS-13) and Rank Students by Bid Credits (AEBS-12) were assigned 5 story points to reflect their higher algorithmic complexity, since they involve processing multiple student bids, seat constraints, and ranking logic simultaneously.

2. Was your backlog well-prioritized?

Ans. Yes. The backlog prioritized the core bidding and allocation workflow – Submit Elective Preferences, Validate Bidding Credits, Validate Prerequisites, and the full allocation engine (AEBS-12 to AEBS-14) – as High priority. This ensured that the essential end-to-end bidding-to-allocation flow was built first, before moving on to secondary features such as notifications and allocation detail display (AEBS-16, AEBS-17), which were correctly marked Medium priority.

3. How did your simulated sprint align with your plan?

Ans. The simulated sprints aligned closely with the release plan. The 15 user stories were split across three one-week sprints of 22, 17, and 11 story points respectively, moving from core bidding capture, to the allocation engine, to override and audit capabilities. Each sprint's work items were progressed through To Do → In Progress → Done on the board, and Sprint 3 closed with all committed points completed, demonstrating realistic scope selection for a one-week timebox.

4. What insights did the burndown chart give about your team's capacity?

Ans. The Burndown Chart for AEBS Sprint 3 showed a steady downward progression from 10 remaining points to 0, tracking close to the ideal guideline rather than dropping in one sharp step. This indicates that the work items (AEBS-18, AEBS-19, AEBS-20) were reasonably well broken down and estimated, and that the team's velocity of roughly 11 points per one-week sprint is a realistic baseline for planning future sprints, such as adding reporting or analytics features on top of the AEBS allocation engine.